COMP9602: Optimization

Convex Programs (I)

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Where we are

Theory	convex set convex function standard forms of optimization problems, quasi-convex optimization linear program quadratic program geometric program vector optimization integer program Duality
Algorithm	unconstrained optimization equality constrained optimization interior-point method localization methods subgradient method decomposition methods etc.

Optimization problem in standard form

Standard form

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

- $x \in \mathbf{R}^n$ is the optimization variable
- $f_0 : \mathbf{R}^n \to \mathbf{R}$ is the objective or cost function
- $f_i : \mathbf{R}^n \to \mathbf{R}$, $i = 1, \ldots, m$, are the inequality constraint functions
- $h_i : \mathbf{R}^n \to \mathbf{R}$ are the equality constraint functions

optimal value:

$$p^\star = \inf\{f_0(x) \mid f_i(x) \leq 0,\ i = 1, \ldots, m,\ h_i(x) = 0,\ i = 1, \ldots, p\}$$

- $p^\star = \infty$ if problem is infeasible (no x satisfies the constraints)
- $p^\star = -\infty$ if problem is unbounded below

Feasible and optimal points

x is **feasible** if $x \in \mathbf{dom}\, f_0$ and it satisfies the constraints

optimal point: a feasible x is **optimal** if $f_0(x) = p^\star$

x is **locally optimal** if there is an $R > 0$ such that x is optimal for

$$\begin{array}{ll} \text{minimize (over } z) & f_0(z) \\ \text{subject to} & f_i(z) \leq 0, \quad i = 1, \ldots, m, \quad h_i(z) = 0, \quad i = 1, \ldots, p \\ & \|z - x\|_2 \leq R \end{array}$$

Examples:

- $f_0(x) = 1/x$, $\mathbf{dom}\, f_0 = \mathbf{R}_{++}$
- $f_0(x) = -\log x$, $\mathbf{dom}\, f_0 = \mathbf{R}_{++}$
- $f_0(x) = x \log x$, $\mathbf{dom}\, f_0 = \mathbf{R}_{++}$

Feasibility problem

$$\begin{array}{ll} \text{find} & x \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

can be considered a special case of the general problem with $f_0(x) = 0$:

$$\begin{array}{ll} \text{minimize} & 0 \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

Convex optimization problem

Standard form convex optimization problem

$$\begin{array}{lll} \text{minimize} & f_0(x) & \\ \text{subject to} & f_i(x) \leq 0, & i = 1, \ldots, m \\ & a_i^T x = b_i, & i = 1, \ldots, p \end{array} \quad \Longleftrightarrow \quad \begin{array}{lll} \text{minimize} & f_0(x) & \\ \text{subject to} & f_i(x) \leq 0, & i = 1, \ldots, m \\ & Ax = b & \end{array}$$

$f_0, f_1, \ldots, f_m$ are convex; equality constraints are affine

important property: feasible set of a convex optimization problem is convex

Generalized inequalities are also ok

- e.g.,

$$\min \mathbf{c}^T \mathbf{x}$$

subject to:

$$A_0 + A_1 x_1 + A_2 x_2 + \ldots + A_n x_n \preceq 0$$

Local and global optima

- Any locally optimal point of a convex program is (globally) optimal

  - proof

Suppose that x is local minimum, but $\exists y \in C(\text{feasibility set}), f_0(y) < f_0(x)$

Consider $z = \theta y + (1-\theta)x$ with small θ (such that z is close to x)

Since feasibility set C is convex, we have $z \in C$

and $f_0(z) = f_0(\theta y + (1-\theta)x) \leq \theta f_0(y) + (1-\theta) f_0(x)$

$< f_0(x)$

This contradicts with x being the local minimum.

[Geometric interpretation]

Optimality criteria for differentiable convex f_0

x is optimal if and only if it is feasible and

$$\nabla f_0(x)^T(y-x) \geq 0 \quad \text{for all feasible } y$$

[Geometric interpretation]

- proof

$\Rightarrow$ Since $f_0(x)$ is convex and differentiable, we have $f_0(y) \geq f_0(x) + \nabla f_0(x)^T(y-x)$

If $\nabla f_0(x)^T(y-x) \geq 0$, then $f_0(y) \geq f_0(x), \forall \text{feasible } y \in C$

$\Leftarrow$ If $f_0(x)$ is minimum and $\nabla f_0(x)^T(y-x) < 0, \text{ for some } y \in C$

Consider $z = \theta y + (1-\theta)x \in C$

$$f_0(z) = f_0(\theta y + (1-\theta)x)$$

$$\left.\frac{\partial f_0(z)}{\partial \theta}\right|_{\theta \to 0} = \nabla f_0(x)^T(y-x) < 0$$

definition of directional derivative

directional derivative of f at x in the direction v:

$$\left.\frac{d}{dt} f(x+tv)\right|_{t=0} = \nabla f(x)^T v$$

For small θ, $f_0(z) < f_0(x)$ which is a contradiction

Optimality criteria for differentiable convex f_0 (cont'd)

- **unconstrained problem**: x is optimal if and only if

$$x \in \mathbf{dom}\, f_0, \qquad \nabla f_0(x) = 0$$

- proof

$\Leftarrow$ Suppose x is minimum. Then $x \in dom f_0, \nabla f_0(x)^T (y - x) \geq 0, \forall y \in dom f_0$

Since f_0 is differentiable, $dom f_0$ is by definition open. So all y sufficiently close to x is feasible.

Let $y = x - \theta \nabla f_0(x), \theta > 0$

We have $\nabla f_0(x)^T (y - x) = -\theta \nabla f_0(x)^T \nabla f_0(x) = -\theta ||\nabla f_0(x)||_2^2 \geq 0$

Therefore $\nabla f_0(x) = 0$

Optimality criteria for differentiable convex f_0 (cont'd)

- **equality constrained problem**

$$\text{minimize} \quad f_0(x) \quad \text{subject to} \quad Ax = b$$

x is optimal if and only if there exists a ν such that

$$x \in \mathbf{dom}\, f_0, \qquad Ax = b, \qquad \nabla f_0(x) + A^T\nu = 0$$

proof:

$\Leftarrow$ We have $\nabla f_0(x)^T(y-x) \geq 0, \forall y$ satisfying $Ay = b$

null space of matrix A

Since x is feasible, every feasible $y = x + \mu$ for some $\mu \in N(A)$

$\nabla f_0(x)^T \mu \geq 0, \forall \mu \in N(A)$

Since $\nabla f_0(x)^T(-\mu) \geq 0$ too, we have $\nabla f_0(x)^T \mu = 0, \forall \mu \in N(A)$

Therefore $\nabla f_0(x) \perp N(A)$

range of matrix A^T

Since $N(A)^\perp = R(A^T)$, we have $\nabla f_0(x) \in R(A^T)$

That is, $\exists -\nu \in \mathbf{R}^p$ such that $A^T(-\nu) = \nabla f_0(x)$

Optimality criteria for differentiable convex f_0 (cont'd)

- **minimization over nonnegative orthant**

$$\text{minimize} \quad f_0(x) \quad \text{subject to} \quad x \succeq 0$$

x is optimal if and only if

$$x \in \mathbf{dom}\, f_0, \qquad x \succeq 0, \qquad \begin{cases} \nabla f_0(x)_i \geq 0 & x_i = 0 \\ \nabla f_0(x)_i = 0 & x_i > 0 \end{cases}$$

proof:

$\Leftarrow \quad x \succeq 0, \nabla f_0(x)^T (y - x) \geq 0, \forall y \succeq 0$

$\nabla f_0(x)^T y$ (consider it as a linear function of y) is unbounded below on $y \succeq 0$ unless $\nabla f_0(x) \succeq 0$

But $-\nabla f_0(x)^T x \geq 0$

So $\nabla f_0(x)^T x = 0$ i.e., $\sum_{i=1}^{n} (\nabla f_0(x))_i x_i = 0$

Since $x_i \geq 0$ and $(\nabla f_0(x))_i \geq 0$, we obtain $(\nabla f_0(x))_i x_i = 0, \forall i$

Equivalent convex programs

two problems are (informally) **equivalent** if the solution of one is readily obtained from the solution of the other, and vice-versa

- **eliminating equality constraints**

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

find

- x_0: a particular solution of Ax=b
- matrix F: whose range is the null space of A, i.e. R(F)=N(A)

is equivalent to

$$\begin{array}{ll} \text{minimize (over } z) & f_0(Fz + x_0) \\ \text{subject to} & f_i(Fz + x_0) \leq 0, \quad i = 1, \ldots, m \end{array}$$

Equivalent convex programs

- **introducing equality constraints**

$$\begin{array}{ll} \text{minimize} & f_0(A_0x + b_0) \\ \text{subject to} & f_i(A_ix + b_i) \leq 0, \quad i = 1, \ldots, m \end{array}$$

is equivalent to

$$\begin{array}{ll} \text{minimize (over } x, \, y_i) & f_0(y_0) \\ \text{subject to} & f_i(y_i) \leq 0, \quad i = 1, \ldots, m \\ & y_i = A_ix + b_i, \quad i = 0, 1, \ldots, m \end{array}$$

Equivalent convex programs

- **introducing slack variables for linear inequalities**

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \ldots, m \end{array}$$

is equivalent to

$$\begin{array}{ll} \text{minimize (over } x,\ s) & f_0(x) \\ \text{subject to} & a_i^T x + s_i = b_i, \quad i = 1, \ldots, m \\ & s_i \geq 0, \quad i = 1, \ldots m \end{array}$$

Equivalent convex programs

- **epigraph form**: standard form convex problem is equivalent to

$$\begin{array}{ll} \text{minimize (over } x,\, t) & t \\ \text{subject to} & f_0(x) \leq t \\ & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

- **minimizing over some variables**

$$\begin{array}{ll} \text{minimize} & f_0(x_1, x_2) \\ \text{subject to} & f_i(x_1) \leq 0, \quad i = 1, \ldots, m \end{array}$$

is equivalent to

$$\begin{array}{ll} \text{minimize} & \tilde{f}_0(x_1) \\ \text{subject to} & f_i(x_1) \leq 0, \quad i = 1, \ldots, m \end{array}$$

where $\tilde{f}_0(x_1) = \inf_{x_2} f_0(x_1, x_2)$

- Small perturbation of the problem makes it very hard (potentially)
  - max convex or minimize concave
  - non-convex constraints
  - convex equality constraints

Quasiconvex optimization

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

with $f_0 : \mathbf{R}^n \to \mathbf{R}$ quasiconvex, $f_1, \ldots, f_m$ convex

can have locally optimal points that are not (globally) optimal

- example

Quasiconvex optimization

- Quasiconvex function $\mathrm{f0(x)}$ can be represented by a family of convex functions $\phi_t(x)$, indexed by $t \in R$

  if f_0 is quasiconvex, there exists a family of functions ϕ_t such that:

  - $\phi_t(x)$ is convex in x for fixed t
  - t-sublevel set of f_0 is 0-sublevel set of ϕ_t, *i.e.*, $f_0(x) \leq t \iff \phi_t(x) \leq 0$

Quasiconvex optimization

- Such a representation always exists and not unique

e.g.:

(1) $\phi_t(x) = \begin{cases} 0, f_0(x) \leq t, \\ \infty, otherwise, \end{cases}$

(2) $\phi_t(x) = dist(x, \{z | f_0(z) \leq t\})$

(3) Convex over concave functions:

$$f_0(x) = \frac{p(x)}{q(x)}$$

with p convex, q concave, and $q(x) > 0$ on $\mathbf{dom}\, f_0$

can take $\phi_t(x) = p(x) - tq(x)$:

- for $t \geq 0$, ϕ_t convex in x
- $p(x)/q(x) \leq t$ if and only if $\phi_t(x) \leq 0$

Quasiconvex optimization via convex feasibility problems

Let p* denote the optimal value of the quasiconvex program

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

Solve the feasibility problem:

$$\begin{array}{ll} \text{find} & x \\ \text{subject to:} & \phi_t(x) \leq 0 \\ & f_i(x) \leq 0, \quad i = 1, \ldots, m, \\ & Ax = b \end{array}$$

if feasible, $p^* \leq t$; otherwise, $p^* > t$

=> solve the quasiconvex problem using bisection, solving a convex feasibility problem at each step

Quasiconvex optimization via convex feasibility problems

Start with an interval [l,u] known to contain p*

Bisection method for quasiconvex optimization

given $l \leq p^\star$, $u \geq p^\star$, tolerance $\epsilon > 0$.

repeat

1. $t := (l+u)/2$.
2. Solve the convex feasibility problem (1).
3. **if** (1) is feasible, $u := t$; **else** $l := t$.

until $u - l \leq \epsilon$.

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requires exactly $\lceil \log_2((u-l)/\epsilon) \rceil$ iterations

- Reference
  - Chapter 4.1– 4.2, Convex Optimization.
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